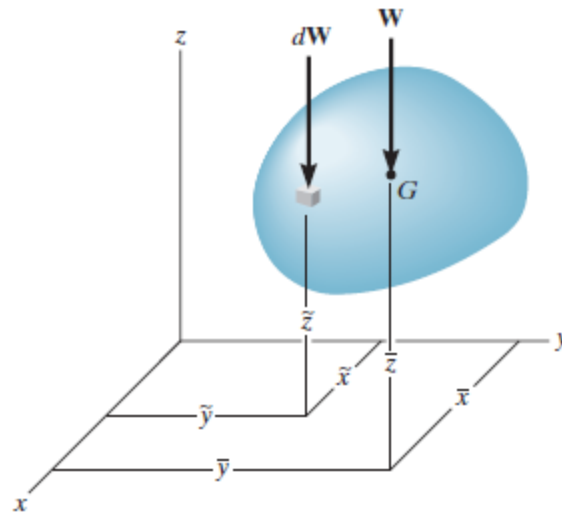


Súlypont

Súlypont

Fontos ismerni a test teljes súlyát, illetve annak elhelyezését, ha elemezni szeretnénk a súly hatását a testre.

Azt a pontot, melybe a test teljes súlyát elhelyezhetjük, súlypontnak nevezzük (Center of gravity). Ez a pont független a kiválasztott koordináta-rendszer-től.



A testet végtelenül sok differenciális méretű (és súlyú) részecske halmazként elemezzük. Ha a test a gravitációs térben helyezkedik el (számunkra a gravitációs tér a test valamennyi pontjára tekintve állandó értékű és azonos irányú-párhuzamos), akkor a részecskék súlyai párhuzamos erőrendszert alkotnak. Az erőrendszer eredője:

$$+\downarrow F_R = \Sigma F_z;$$

(a felső képre érvényes, de ha például az x tengely körül elforgatnánk az koordinátarendszert-testet, akkor a súlyerők az y irányban lennének...)

$$W = \int dW$$

A súlypont elhelyezése a koordinátarendszer tengelyeihez képest úgy határozható meg, hogy kiegyenlítjük a részecskék súlyainak teljes forgató hatását az adott tengelyhez (tengelyekhez) képest, a test teljes súlyának forgató hatásával az adott tengelyhez (tengelyekhez) képest.

$$(M_R)_y = \Sigma M_y;$$

(a képlet az y tengelyre vonatkozik)

$$\bar{x}W = \int \tilde{x}dW$$

$$\bar{x} = \frac{\int \tilde{x} dW}{\int dW} \quad \bar{y} = \frac{\int \tilde{y} dW}{\int dW} \quad \bar{z} = \frac{\int \tilde{z} dW}{\int dW} \quad (9-1)$$

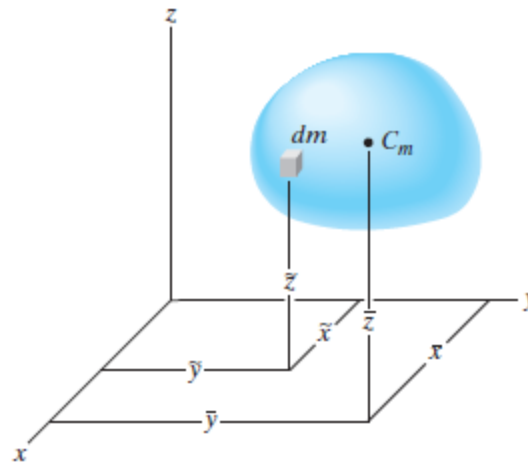
where

$\bar{x}, \bar{y}, \bar{z}$ are the coordinates of the center of gravity G .

$\tilde{x}, \tilde{y}, \tilde{z}$ are the coordinates of an arbitrary particle in the body.

Tömegközpont

Ha elemezni akarjuk a testek mozgását valamilyen erők hatása eredményeként, fontos ismerni a test tömegközpontját (Center of Mass of a body).

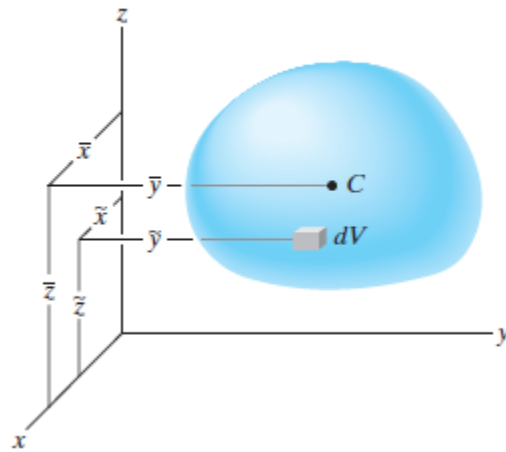


$$dW = g \, dm$$

$$\bar{x} = \frac{\int \tilde{x} \, dm}{\int dm} \quad \bar{y} = \frac{\int \tilde{y} \, dm}{\int dm} \quad \bar{z} = \frac{\int \tilde{z} \, dm}{\int dm}$$

Geometriai központ (Centroid)

Térfogat középpontja.



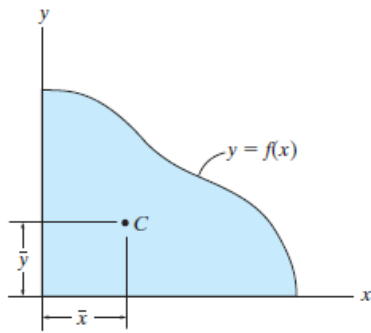
Ha a test homogén, felírható:

$$dm = \rho dV.$$

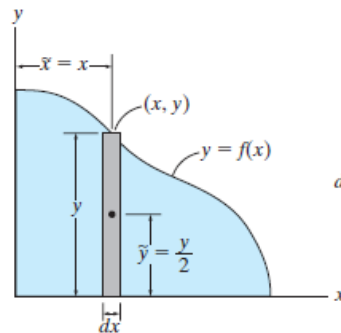
Ekkor:

$$\bar{x} = \frac{\int_V \tilde{x} dV}{\int_V dV} \quad \bar{y} = \frac{\int_V \tilde{y} dV}{\int_V dV} \quad \bar{z} = \frac{\int_V \tilde{z} dV}{\int_V dV}$$

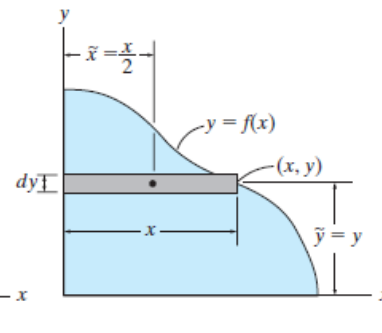
Sík középpontja.



(a)



(b)



(c)

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA} \quad \bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

$$y = f(x).$$

Ha a differenciális felülethez a fenti ábra második vagy harmadik képen bemutatott felületeket vesszük, akkor egyszeres integrálást végezhetünk.

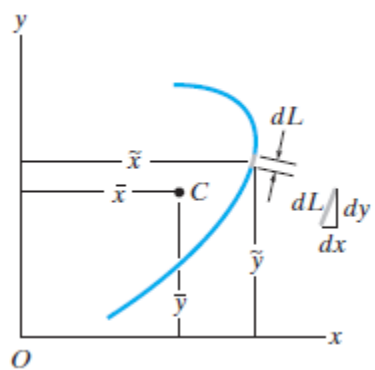
$$dA = y dx, \quad \tilde{x} = x, \quad \tilde{y} = y/2,$$

vagy

a c ábrát nézve

$$dA = x dy, \quad \tilde{x} = x/2, \quad \tilde{y} = y$$

Síkgörbe középpontja



Ha síkban vagyunk:

$$y = f(x).$$

$$\bar{x} = \frac{\int_L \tilde{x} dL}{\int_L dL} \quad \bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL}$$

$$dL = \sqrt{(dx)^2 + (dy)^2}$$

$$\begin{aligned}
 dL &= \sqrt{\left(\frac{dx}{dy}\right)^2 dy^2 + \left(\frac{dy}{dx}\right)^2 dx^2} \\
 &= \left(\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\right) dx
 \end{aligned}$$

or

$$\begin{aligned}
 dL &= \sqrt{\left(\frac{dx}{dy}\right)^2 dy^2 + \left(\frac{dy}{dx}\right)^2 dy^2} \\
 &= \left(\sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\right) dy
 \end{aligned}$$

Azt a változatot alkalmazzuk, amelyik egyszerűbb integráláshoz vezet.

EXAMPLE 9.1

Locate the centroid of the rod bent into the shape of a parabolic arc as shown in Fig. 9–8.

SOLUTION

Differential Element. The differential element is shown in Fig. 9–8. It is located on the curve at the *arbitrary point* (x, y) .

Area and Moment Arms. The differential element of length dL can be expressed in terms of the differentials dx and dy using the Pythagorean theorem.

$$dL = \sqrt{(dx)^2 + (dy)^2} = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} dy$$

Since $x = y^2$, then $dx/dy = 2y$. Therefore, expressing dL in terms of y and dy , we have

$$dL = \sqrt{(2y)^2 + 1} dy$$

As shown in Fig. 9–8, the centroid of the element is located at $\tilde{x} = x$, $\tilde{y} = y$.

Integrations. Applying Eq. 9–5 and using the integration formula to evaluate the integrals, we get

$$\begin{aligned} \bar{x} &= \frac{\int_L \tilde{x} dL}{\int_L dL} = \frac{\int_0^{1\text{ m}} x \sqrt{4y^2 + 1} dy}{\int_0^{1\text{ m}} \sqrt{4y^2 + 1} dy} = \frac{\int_0^{1\text{ m}} y^2 \sqrt{4y^2 + 1} dy}{\int_0^{1\text{ m}} \sqrt{4y^2 + 1} dy} \\ &= \frac{0.6063}{1.479} = 0.410 \text{ m} \quad \text{Ans.} \end{aligned}$$

$$\bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL} = \frac{\int_0^{1\text{ m}} y \sqrt{4y^2 + 1} dy}{\int_0^{1\text{ m}} \sqrt{4y^2 + 1} dy} = \frac{0.8484}{1.479} = 0.574 \text{ m} \quad \text{Ans.}$$

NOTE: These results for C seem reasonable when they are plotted on Fig. 9–8.

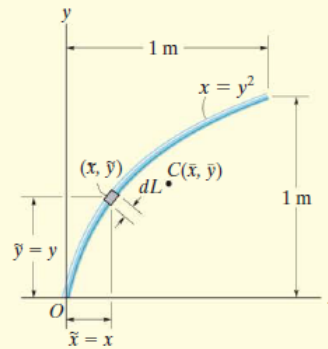


Fig. 9–8

EXAMPLE 9.2

Locate the centroid of the circular wire segment shown in Fig. 9–9.

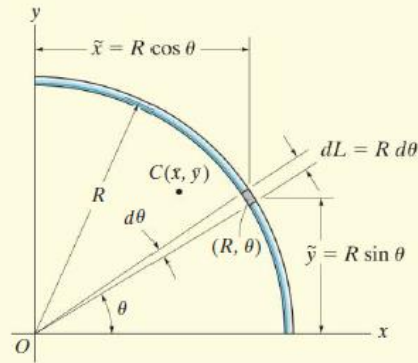


Fig. 9–9

SOLUTION

Polar coordinates will be used to solve this problem since the arc is circular.

Differential Element. A differential circular arc is selected as shown in the figure. This element lies on the curve at (R, θ) .

Length and Moment Arm. The length of the differential element is $dL = R d\theta$, and its centroid is located at $\tilde{x} = R \cos \theta$ and $\tilde{y} = R \sin \theta$.

Integrations. Applying Eqs. 9–5 and integrating with respect to θ , we obtain

$$\bar{x} = \frac{\int_L \tilde{x} dL}{\int_L dL} = \frac{\int_0^{\pi/2} (R \cos \theta) R d\theta}{\int_0^{\pi/2} R d\theta} = \frac{R^2 \int_0^{\pi/2} \cos \theta d\theta}{R \int_0^{\pi/2} d\theta} = \frac{2R}{\pi} \quad \text{Ans.}$$

$$\bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL} = \frac{\int_0^{\pi/2} (R \sin \theta) R d\theta}{\int_0^{\pi/2} R d\theta} = \frac{R^2 \int_0^{\pi/2} \sin \theta d\theta}{R \int_0^{\pi/2} d\theta} = \frac{2R}{\pi} \quad \text{Ans.}$$

NOTE: As expected, the two coordinates are numerically the same due to the symmetry of the wire.

Példa – háromszög súlypontja

EXAMPLE 9.3

Determine the distance $\bar{y}$ measured from the x axis to the centroid of the area of the triangle shown in Fig. 9-10.

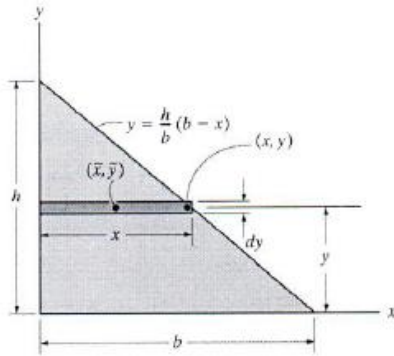


Fig. 9-10

SOLUTION

Differential Element. Consider a rectangular element having a thickness dy , and located in an arbitrary position so that it intersects the boundary at (x, y) , Fig. 9-10.

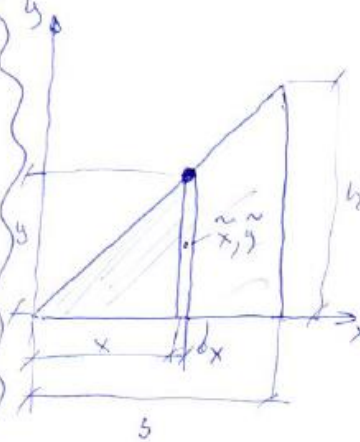
Area and Moment Arms. The area of the element is $dA = x dy$, and its centroid is located a distance $\bar{y} = y$ from the x axis.

Integration. Applying the second of Eqs. 9-4 and integrating with respect to y yields

$$\bar{y} = \frac{\int_A \bar{y} dA}{\int_A dA} = \frac{\int_0^h y \left[\frac{b}{h}(h - y) dy \right]}{\int_0^h \frac{b}{h}(h - y) dy} = \frac{\frac{1}{6}bh^2}{\frac{1}{2}bh} = \frac{h}{3}$$

Ans.

NOTE: This result is valid for any shape of triangle. It states that the centroid is located at one-third the height, measured from the base of the triangle.



$$\frac{h}{b} = \frac{y}{x} \Rightarrow$$

$$y(x) = x \frac{h}{b}$$

$$\begin{aligned} dA &= y \cdot dx = x \frac{h}{b} dx \\ \bar{x} &= \frac{\int_A \bar{x} dA}{\int_A dA} = \frac{\int_0^b x \cdot x \frac{h}{b} dx}{\int_0^b x \frac{h}{b} dx} \\ &= \frac{\int_0^b \frac{x^2}{3} dx}{\int_0^b \frac{x}{2} dx} = \frac{\left[\frac{x^3}{9} \right]_0^b}{\left[\frac{x^2}{4} \right]_0^b} = \frac{\frac{b^3}{9}}{\frac{b^2}{4}} = \frac{4b}{9} \end{aligned}$$

EXAMPLE 9.4

Locate the centroid for the area of a quarter circle shown in Fig. 9–11.

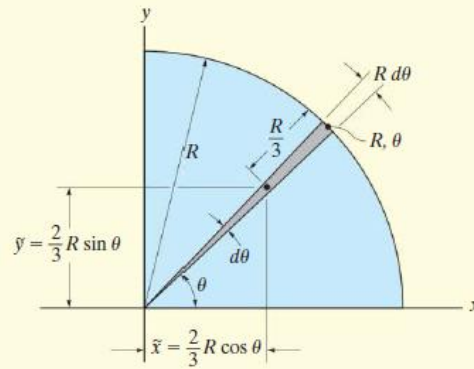


Fig. 9–11

SOLUTION

Differential Element. Polar coordinates will be used, since the boundary is circular. We choose the element in the shape of a *triangle*, Fig. 9–11. (Actually the shape is a circular sector; however, neglecting higher-order differentials, the element becomes triangular.) The element intersects the curve at point (R, θ) .

Area and Moment Arms. The area of the element is

$$dA = \frac{1}{2}(R)(R d\theta) = \frac{R^2}{2}d\theta$$

and using the results of Example 9.3, the centroid of the (triangular) element is located at $\tilde{x} = \frac{2}{3}R \cos \theta$, $\tilde{y} = \frac{2}{3}R \sin \theta$.

Integrations. Applying Eqs. 9–4 and integrating with respect to θ , we obtain

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA} = \frac{\int_0^{\pi/2} \left(\frac{2}{3}R \cos \theta \right) \frac{R^2}{2} d\theta}{\int_0^{\pi/2} \frac{R^2}{2} d\theta} = \frac{\left(\frac{2}{3}R \right) \int_0^{\pi/2} \cos \theta d\theta}{\int_0^{\pi/2} d\theta} = \frac{4R}{3\pi} \quad \text{Ans.}$$

$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA} = \frac{\int_0^{\pi/2} \left(\frac{2}{3}R \sin \theta \right) \frac{R^2}{2} d\theta}{\int_0^{\pi/2} \frac{R^2}{2} d\theta} = \frac{\left(\frac{2}{3}R \right) \int_0^{\pi/2} \sin \theta d\theta}{\int_0^{\pi/2} d\theta} = \frac{4R}{3\pi} \quad \text{Ans.}$$

Összetett testek súlypontja

Az összetett testet több egyszerű test összességéként lehet értelmezni, ahol az egyszerűbb testek súlypontját ismerjük (háromszög, négyzög, kör...).

Ilyenkor a következő az eljárás:

$$\bar{x} = \frac{\sum \tilde{x}W}{\sum W} \quad \bar{y} = \frac{\sum \tilde{y}W}{\sum W} \quad \bar{z} = \frac{\sum \tilde{z}W}{\sum W} \quad ($$

Here

$\bar{x}, \bar{y}, \bar{z}$ represent the coordinates of the center of gravity G of the composite body.

$\tilde{x}, \tilde{y}, \tilde{z}$ represent the coordinates of the center of gravity of each composite part of the body.

$\sum W$ is the sum of the weights of all the composite parts of the body, or simply the total weight of the body.

Ha esetleg a test üreges, ezeket az üregeket „negatív súlyú testként” vesszük figyelembe.

Az adatokat rendszerint táblázat formában rendezzük, hogy egyszerűbb legyen az elemzés.

EXAMPLE 9.9

Locate the centroid of the wire shown in Fig. 9-16a.

SOLUTION

Composite Parts. The wire is divided into three segments as shown in Fig. 9-16b.

Moment Arms. The location of the centroid for each segment is determined and indicated in the figure. In particular, the centroid of segment ① is determined either by integration or by using the table on the inside back cover.

Summations. For convenience, the calculations can be tabulated as follows:

Segment	L (mm)	$\tilde{x}$ (mm)	$\tilde{y}$ (mm)	$\tilde{z}$ (mm)	$\tilde{x}L$ (mm ²)	$\tilde{y}L$ (mm ²)	$\tilde{z}L$ (mm ²)
1	$\pi(60) = 188.5$	60	-38.2	0	11 310	-7200	0
2	40	0	20	0	0	800	0
3	20	0	40	-10	0	800	-200
	$\Sigma L = 248.5$				$\Sigma \tilde{x}L = 11\,310$	$\Sigma \tilde{y}L = -5600$	$\Sigma \tilde{z}L = -200$

Thus,

$$\bar{x} = \frac{\Sigma \tilde{x}L}{\Sigma L} = \frac{11\,310}{248.5} = 45.5 \text{ mm} \quad \text{Ans.}$$

$$\bar{y} = \frac{\Sigma \tilde{y}L}{\Sigma L} = \frac{-5600}{248.5} = -22.5 \text{ mm} \quad \text{Ans.}$$

$$\bar{z} = \frac{\Sigma \tilde{z}L}{\Sigma L} = \frac{-200}{248.5} = -0.805 \text{ mm} \quad \text{Ans.}$$

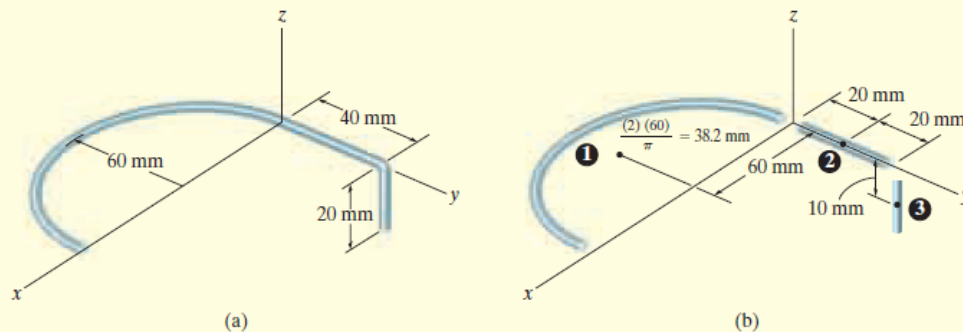


Fig. 9-16

EXAMPLE 9.10

Locate the centroid of the plate area shown in Fig. 9-17a.

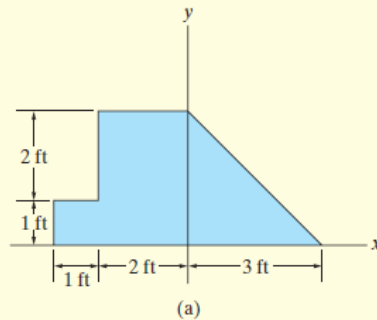


Fig. 9-17

SOLUTION

Composite Parts. The plate is divided into three segments as shown in Fig. 9-17b. Here the area of the small rectangle (3) is considered “negative” since it must be subtracted from the larger one (2).

Moment Arms. The centroid of each segment is located as indicated in the figure. Note that the $\tilde{x}$ coordinates of (2) and (3) are *negative*.

Summations. Taking the data from Fig. 9-17b, the calculations are tabulated as follows:

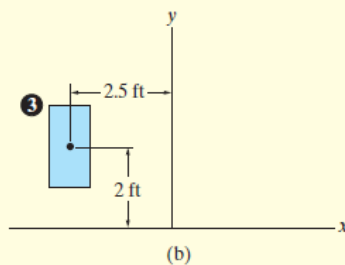
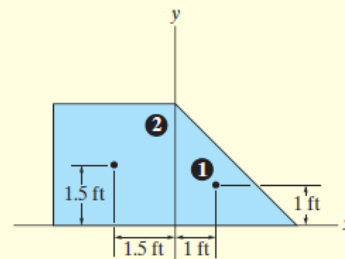
Segment	$A \text{ (ft}^2\text{)}$	$\tilde{x} \text{ (ft)}$	$\tilde{y} \text{ (ft)}$	$\tilde{x}A \text{ (ft}^3\text{)}$	$\tilde{y}A \text{ (ft}^3\text{)}$
1	$\frac{1}{2}(3)(3) = 4.5$	1	1	4.5	4.5
2	$(3)(3) = 9$	-1.5	1.5	-13.5	13.5
3	$-(2)(1) = -2$	-2.5	2	5	-4
	$\Sigma A = 11.5$			$\Sigma \tilde{x}A = -4$	$\Sigma \tilde{y}A = 14$

Thus,

$$\bar{x} = \frac{\Sigma \tilde{x}A}{\Sigma A} = \frac{-4}{11.5} = -0.348 \text{ ft} \quad \text{Ans.}$$

$$\bar{y} = \frac{\Sigma \tilde{y}A}{\Sigma A} = \frac{14}{11.5} = 1.22 \text{ ft} \quad \text{Ans.}$$

NOTE: If these results are plotted in Fig. 9-17a, the location of point *C* seems reasonable.

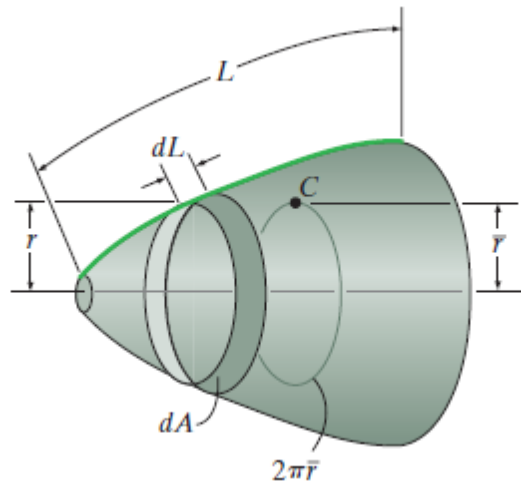


Pappus – Guldinus tételek

A tételek alkalmazásával a forgástestek felületét és térfogatát lehet meghatározni.

Testek felülete

Ha egy síkbeli görbét egy tengely körül elforgatunk (a tengely és a görbe nem metszik egymást), a görbe egy felületet hoz létre.



$$dA = 2\pi r * dL$$

$$A = 2\pi \int r dL$$

Másrészt:

$$\int r dL = \bar{r}L$$

tehát

$$A = 2\pi \bar{r}L$$

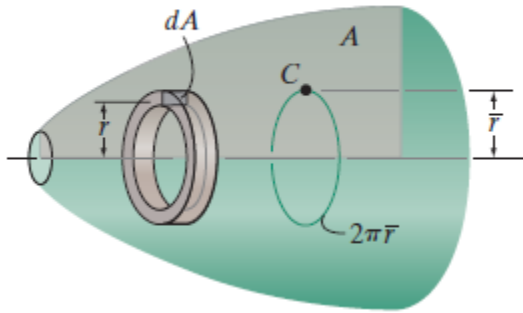
vagyis ha az elforgatás egy θ szögben jön létre:

$$A = \theta \bar{r}L$$

Tehát az első tétel azt mondja ki, hogy a felület nagysága nem más mind a görbe hossza beszorozva a hosszal (körív) amit a görbe súlypontja ír le az elforgatás során.

Testek térfogata

Ha egy sík felületet egy tengely körül elforgatunk (a tengely és a felület nem metszik egymást), a felület egy térfogatot hoz létre.



$$dV = 2\pi r * dA$$

$$V = 2\pi \int r dA$$

Másrészt:

$$\int r dA = \bar{r}A$$

tehát

$$V = 2\pi\bar{r}A$$

vagyis ha az elforgatás egy θ szögben jön létre:

$$V = \theta\bar{r}A$$

Tehát a második tétel azt mondja ki, hogy a térfogat nagysága nem más mind a felület beszorozva a hosszal (körív) amit a felület súlypontja ír le az elforgatás során.

EXAMPLE 9.13

Determine the surface area and volume of the full solid in Fig. 9-22a.

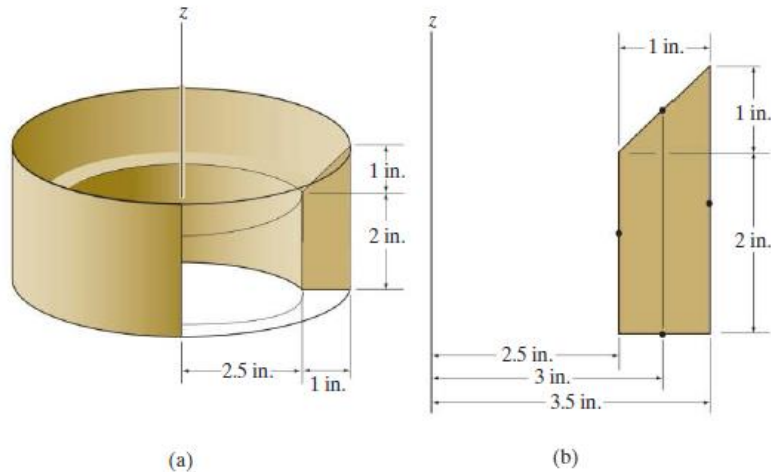


Fig. 9-22

SOLUTION

Surface Area. The surface area is generated by revolving the four line segments shown in Fig. 9-22b 2π radians about the z axis. The distances from the centroid of each segment to the z axis are also shown in the figure. Applying Eq. 9-7 yields

$$\begin{aligned}
 A &= 2\pi \sum \bar{r}L = 2\pi[(2.5 \text{ in.})(2 \text{ in.}) + (3 \text{ in.})\left(\sqrt{(1 \text{ in.})^2 + (1 \text{ in.})^2}\right) \\
 &\quad + (3.5 \text{ in.})(3 \text{ in.}) + (3 \text{ in.})(1 \text{ in.})] \\
 &= 143 \text{ in}^2
 \end{aligned}$$

Ans.

Volume. The volume of the solid is generated by revolving the two area segments shown in Fig. 9-22c 2π radians about the z axis. The distances from the centroid of each segment to the z axis are also shown in the figure. Applying Eq. 9-10, we have

$$\begin{aligned}
 V &= 2\pi \sum \bar{r}A = 2\pi \left\{ (3.1667 \text{ in.}) \left[\frac{1}{2} (1 \text{ in.})(1 \text{ in.}) \right] + (3 \text{ in.})[(2 \text{ in.})(1 \text{ in.})] \right\} \\
 &= 47.6 \text{ in}^3
 \end{aligned}$$

Ans.

